

Aditya Hansoty

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EDUCATION

University of Michigan

B.S.E. Biomedical Engineering, Minor in Computer Science

- GPA: 3.73/4.0

Ann Arbor, MI

Aug. 2025 – May 2029

EXPERIENCE

Systems Engineering Intern

Naval Medical Research Unit, Department of the Navy

- Processing physiological signals and applying deep learning to support medical device development for military medicine

Oct. 2026 – Dec. 2026

San Antonio, TX

Prosthetics Research Intern

Rehabilitation Robotics Lab, University of Illinois Chicago

- Designed and iterated threaded encoder caps for a powered ankle-foot prosthesis in SolidWorks, progressing through 3D-printed and CNC-machined revisions to improve mechanical fit
- Controlled a powered ankle-foot prosthesis via TwinCAT and led Vicon motion capture across 10 human-subject trials; built Python pipelines synchronizing EMG, ECG, and metabolic cost data for a manuscript presented at IROS 2026
- Engineered a signal-agnostic AR biofeedback platform (Python, Unity, HoloLens 2) pixel-streaming live via WebSockets for real-time gait feedback; sole developer, open-sourced
- First-author manuscript submitted to ICRA 2027 demonstrating HoloLens 2 latency (120.2 ms) matched a wired monitor (119.2 ms) via OpenCV tracking; 2-subject pilot showed AR guidance reduced step-length asymmetry by 41% and 53%

May 2026 – Sept. 2026

Chicago, IL

Exoskeleton Research Intern

Locomotor Control Systems Laboratory, University of Michigan

- Collecting and processing motion-capture and EMG data from clinical trials evaluating a wearable knee exoskeleton in patients with osteoarthritis and stroke
- Redesigning hip-belt geometry and dorsal motor-housing mounts in SolidWorks to improve wearability and reduce sag

Sept. 2026 – Present

Ann Arbor, MI

Cancer Biology Research Intern

University of Chicago Medicine

- Investigated β -tubulin isotype knockouts in bone-metastatic breast cancer cells using cell culture, adhesion assays, Western blotting, and confocal microscopy; presented findings to 100+ researchers at the UChicago Cancer Center Symposium

June 2025 – Aug. 2025

Chicago, IL

PROJECTS

Tibial Locking Compression Plate | SolidWorks, FEA, Simulink

- Designed a parametric 10-hole Ti-6Al-4V locking compression plate in SolidWorks and validated it to ASTM F382 via FEA
- Achieved factors of safety of 1.73 under 500 N four-point bending and 2.36 under 10 N·m torsion; mesh converged within 3%
- Developed a Simulink bone-healing model predicting a 46% reduction in plate stress over 12 weeks

Aug. 2026 – Sept. 2026

EMG-Controlled Prosthetic Gripper | SolidWorks, Arduino, Python

- Designed and 3D-printed a 2-finger prosthetic gripper integrating an MG996R servo, MyoWare 2.0 EMG sensor, and Arduino Uno for real-time gesture-controlled actuation
- Benchmarked two EMG classification pipelines, achieving 99.4% three-gesture classification accuracy using LDA

May 2025 – July 2026

ACTIVITIES

M-HEAL Team PACT

Design Subteam

- Redesigning a urine-collection device in SolidWorks for HPV-based cervical cancer screening in Ghana, adapting geometry for pit and pan latrines and translating clinician and patient feedback into design requirements

Sept. 2026 – Present

Ann Arbor, MI

Biomedical Engineering Society

Professional Development Executive Board

- Led 10+ technical workshops for 100+ members on SolidWorks, wet-lab techniques, and resume development

Aug. 2025 – Present

Ann Arbor, MI

TECHNICAL SKILLS

Engineering: SolidWorks, SolidWorks Simulation (FEA), 3D Printing, CNC Machining, Arduino, Simulink, TwinCAT, Vicon Motion Capture, Signal Processing, Lab Streaming Layer

Programming: Python, C++, MATLAB, Unity (C#)

Data & ML: OpenCV, WebSockets, NumPy, Pandas, SciPy, scikit-learn